State-Level Changes in Community Supervision and Infant Mortality, 2001-2018

AP Pittman[a], Daichi Hibi[b], Michael Cao[a], Isabella Bouklas[a], David Rigby[c], Tim A. Bruckner[d], Allison Stolte[e], Joan A. Casey[f,g], Alison Gemmill[h], Hedwig Lee[a]

[a] *Department of Sociology, Duke University, Durham, NC 27708*
[b] *Department of Sociology, Washington University in St. Louis, St. Louis, MO 63130*
[c] *Institute for Social Research, University of Michigan, Ann Arbor, MI 48106*
[d] *Department of Health, Society, and Behavior, University of California, Irvine, Irvine, CA 92697*
[e] *Department of Sociology, University of Virginia, Charlottesville, VA 22903*
[f] *Department of Environmental and Occupational Health Sciences, University of Washington, Seattle, WA 98105*
[g] *Department of Epidemiology, University of Washington, Seattle, WA 98105*
[h] *Department of Epidemiology, UCLA Fielding School of Public Health, Los Angeles, CA 90095*

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Abstract

Scholars have critiqued American incarceration as a system of racialized social control that disenfranchises the Black population in ways that may have health consequences. Prior work finds that Black-White disparities in state-level imprisonment rates correlate with stark Black-White disparities in infant health. Less is known about the infant health consequences of community supervision (probation and parole), which plays an increasingly central role in the US carceral system. Using novel state-level data (2001 to 2018), we use two-way fixed effects models to examine whether overall and race-specific imprisonment, probation, and parole rates are each associated with race-specific infant mortality rates. We find that different components of the carceral state shape infant health in distinct ways. For example, a one standard-deviation increase in total imprisonment rates corresponds with an increase of 0.65 (*SE: 0.30*) deaths per 1,000 live births among the Black population only. Rates of probation and parole among a state's Black population, however, demonstrate a protective effect on Black infant mortality (0.32 (*SE: 0.12*) and 0.55 (*SE: 0.15*) fewer deaths per 1,000 live births), independent of each other and of imprisonment rates. The White probation rate appears deleterious for White births, but only in states with the highest imprisonment rates, suggesting an interactive effect. These findings indicate that community supervision may operate in a meaningfully different way for Black and White populations and emphasize the need for future work that considers community supervision as an important structural determinant of health.

Keywords: community supervision; probation/parole; infant mortality; racial disparities; population health

Main Text

Introduction and Background

The United States (US) imprisons people at a significantly higher rate than any other high-income country (Garland, 2013), increasing over four-fold between the early 1970s and 2010s (National Research Council, 2014). This increase is particularly troubling since imprisonment is often critiqued as a system of racialized social control, where the national Black imprisonment rate is six times higher than the White imprisonment rate (Alexander, 2012). Yet, prisons are only one site of the larger carceral state. This paper aims to locate an even larger site—community supervision—and to assess the extent to which it merits scholarly attention.

The term "community supervision" refers to probation and parole supervision. Probation, a sentence to correctional supervision outside of custody in prison or jail, and parole, a post-custody provisional supervision, originated as "correctionalist commitments" to rehabilitation and welfare relative to incarceration (Garland, 2002). Community supervision is seen as a more beneficial and salubrious form of control compared to jail or prison incarceration. However, the population-level impacts of community supervision remain understudied, and it remains unclear whether the effects of increased community supervision (as alternative to incarceration) are net-positive or net-negative for health and wellbeing (Phelps & Ruhland, 2022).

Probation and parole capture the vast majority of those under carceral control. More US adults are on probation and parole on any given day than in prison or jail. The latest US estimates report 3.1 million adults on probation and 680,000 adults on parole, compared to 1.3 million in prison and 660,000 in jail (Mueller & Kluckow, 2025; Kaeble, 2025; Zeng, 2025). Rates of community supervision have also increased since the 1970s prison boom (Rigby et al., 2024). Some scholars and criminal justice advocates suggest that the US has reached a new era of mass punishment (Phelps, 2014), which wields incarceration in its right hand and community supervision in its left.[1] Despite these shifts, literature focused on the consequences of the carceral state is often limited to incarceration. Whereas prior work associates incarceration with a range of "collateral consequences" at the individual, state, and national levels (Western & Wildeman, 2009), the potential consequences of community supervision have been largely unexplored. The question remains as to whether community supervision serves as a less harmful alternative to incarceration.

We begin to address this question by examining and comparing the consequences of incarceration and community supervision on population health. Specifically, we focus on infant mortality, a widely accepted measure of health and quality of life in a population

[1] Phelps (2014) advances a compelling argument that after expanding together, probation and incarceration rates decoupled after 1980, and that state supervision rates are now driven by probation, not imprisonment. Moreover, probation varies greatly across states, and may be strategically used to absorb "the punitive excesses of criminalization in contexts where imprisonment rates were tightly managed" (Phelps, 2014, p. 4).

(Reidpath & Allotey, 2003). The infant mortality rate (IMR) is defined as the number of infants that die before their first birthday per 1,000 live births (Vilda et al., 2021), and at a rate of 5.61 deaths per 1,000 live births (Ely & Driscoll, 2024), the US has the highest IMR among high-income countries (Thakrar et al., 2018). Infant mortality rates also reflect large racial inequalities in the US. In 2022, Black infants died more than twice as often as White infants, at a rate of 10.90 to 4.52 deaths per 1,000 live births, respectively (Ely & Driscoll, 2024; Thakrar et al., 2018).

Prior research has shown that incarceration increases infant mortality risk. At the individual level, parental incarceration (of a mother, father, or both) during or directly before the prenatal period predicts increased risk of infant mortality and other adverse birth outcomes (Austin et al., 2022), likely driven by incarceration-related financial stressors, social strains via familial disruption, and other proliferating stressors (Marziali et al., 2024). Similar associations hold for neighborhood- and county-level rates of incarceration and infant mortality (Holaday et al., 2023; McKenna & Nowotny, 2024), with authors suggesting pathways such as drained local economies, strained social networks, and heightened vigilance (Jahn et al., 2020). These health consequences, however, are not equally distributed. Prior work finds that the average state imprisonment rate is associated positively and substantially with the Black IMR, but not with the White IMR (Conway, 2021; Wildeman, 2012), thus contributing to Black-White disparities in infant mortality.

In contrast to the relatively well-documented negative effects of incarceration on health, there is limited evidence of the relation between community supervision and health, especially at the population level. Prior work overwhelmingly focuses on the individual level. Research finds that adults who experienced community supervision have higher risk of chronic and infectious diseases and substance use disorders (Hawks et al., 2022; Winkelman et al., 2020). In fact, those on probation have 2-3 times higher age-adjusted mortality rates than incarcerated individuals (Wildeman et al., 2019). This growing literature importantly connects community supervision to individual health outcomes. However, there is limited understanding of the impact of community supervision on population health, especially within the context of mass incarceration. A population-level examination holds relevance when considering community supervision as a broader policy that reflects societal tolerance of, and treatment for, criminal behavior. In this way, ecological analyses may better capture the full extent of institutions like incarceration and community supervision, which not only affect those directly involved in them, but also are manifestations of structural forces that impact the station of whole communities.

We extend research on the collateral consequences of mass incarceration by examining the relations among imprisonment, probation, parole, and IMR, using novel US state-level data from 2001 through 2018. Drawing from newly available state-level estimates of probation and parole (Rigby et al., 2024), pooled with existing imprisonment data, we create a comprehensive panel of punishment data and use it to predict IMRs across states and years. We aim to assess the extent to which community supervision may work in concert with incarceration to shape population health and health inequalities.

Data and Methods

Data

Data for state-level predictors, outcomes, and covariates were collected from administrative sources. Table 1 displays descriptive statistics, units of measurement, and sources of these data. Our study period is 2001-2018. See Supplementary Information Table S1 for descriptive statistics of all race-specific variables. While our outcome and covariate data span 2002-2018, our predictor data, including imprisonment, probation, and parole, are lagged by one year (2001-2017) to precede the outcome.

[TABLE 1]

We calculate state-level IMR (by birth cohort), imprisonment, probation, and parole rates for non-Hispanic White and non-Hispanic Black populations (throughout the paper, these are sometimes referred to as White and Black or NHW and NHB, respectively). To improve the interpretability of results, we standardize rates of imprisonment, probation, and parole. We suppressed Black IMRs for state-years with fewer than 10 infant deaths, following the US Centers for Disease Control and Prevention guidelines on data reporting. We then included only states with a full panel of non-suppressed infant mortality data (n=34), which is consistent with existing literature examining the links between structural racism and health (see Brown & Homan, 2024). These 34 states account for 92% of the total population, 98% of the Black population, and 91% of the White population, according to 2018 American Community Survey (ACS) estimates (United States Census Bureau, 2018). California, Illinois, and Kansas do not report race-specific rates of imprisonment, probation, or parole, which limits our sample of state-years for race-specific analyses. Missingness on control variables (listed in Table 1) was minimal: two state-years had suppressed values for items that contributed to the high-risk birth index and were excluded from the adjusted analyses. Otherwise, all cases were complete. A map of included states by analysis type can be found in the Supplementary Information.

Analysis

In the first phase of our analysis, we describe state-level trends in IMR, imprisonment, probation, and parole. We also examine state-level variation in the relationship between probation and imprisonment rates and the extent to which they change over time. For this phase of analysis, we use all of the available data, whether or not a state-year was included in subsequent analyses. In the second phase of our analysis, we estimate two-way fixed effects models to examine the independent associations between imprisonment, probation, and parole rates and race-specific IMRs. Two-way fixed effects models simultaneously control for time-invariant differences across units (here, states) *and* unit-invariant differences across time (Wooldridge, 2010).

Based on prior analyses (Wildeman, 2012), we then include a slate of contemporaneous, state-level controls (homicide rates, violent crime rates, gross domestic product (GDP) per capita, per capita state health expenditure, average SNAP benefits per recipient family, unemployment rates, and indices of population heterogeneity, poverty, and high-risk births) in a second model. The inclusion of these covariates balances the need for ruling out spurious relationships with the desire to avoid over-controlling for possible mediators (see Supplementary Information). All final models (both unadjusted and adjusted) include an interaction between imprisonment rates and probation rates, in an effort to operationalize our expectation that these systems may interact to form qualitatively and quantitatively different realities for states that are high in both, neither, or just one of imprisonment or probation (Phelps, 2013, 2017, 2020). We include parole in our models but do not include interaction terms[2], because while parole is a dimension of community supervision, it does not function as an alternative to incarceration. Parole is a conditional release from prison after serving part of a sentence, while probation is a sentence imposed by a judge in lieu of incarceration. In other words, parole is usually conditional on incarceration.

We estimate each of the above models separately for three main outcomes: the Black IMR, the White IMR, and the difference between these two rates (which captures Black-White disparities). Imprisonment, probation, and parole rates reflect the numbers at the *end* of the calendar year in which they were measured, so they are lagged by one year in all analyses in order to assure correct temporal ordering. Preliminary models used a single predictor at a time, along with our slate of controls, and results were substantively similar to those presented here; results from these preliminary models are available in Supplementary Information Tables S5-S7. In our main analysis, we run models using the total (all-race) rates of imprisonment, probation, and parole *and* models using race-specific rates of imprisonment, probation, and parole, both with and without controls. This results in four models estimating the association between criminal-legal supervision and each outcome. We report results for Black and White IMRs in the Results section. Results for the Black-White difference in IMRs are included in Supplementary Information (Table S4) and were substantively similar to the main results for Black IMR.

Results

Descriptive Results

Figure 1 plots states' race-specific IMRs over time. Across the study period, Black IMRs remain universally higher than White IMRs. Although IMRs decrease for all groups, some states show larger improvement than others, with across- and within-state variation in the race-specific rate changes. For example, Wisconsin has a large drop in Black IMR, but a below-average decrease in White IMR. Meanwhile, Delaware

[2] In sensitivity analyses (available upon request), we did include interactions between parole and incarceration; none of these model results showed statistically detectable interaction effects between parole and incarceration, and main effects were consistent with the published models.

experiences substantial reductions in White IMR, but closer-to-average changes in Black IMR.

[FIGURES 1 + 2]

Figure 2 illustrates the relationship between (standardized) imprisonment rates and probation rates across states and populations in 2017. The top panel allows us to identify states that exhibit either high (upper right) or low (lower left) total rates for both imprisonment or probation, or states high on just imprisonment (lower right) or just probation (upper left). There is a spread of states across the quadrants, consistent with theory that different imprisonment levels and probation levels combine to form distinct realities. New Hampshire, for example, is relatively low on both imprisonment rates and probation rates. Georgia is extremely high on probation rates, but closer to average in imprisonment rates; Oklahoma, by contrast, is one of the highest states on imprisonment rates, but average in probation rates.

Similar to the top panel, we see variation in the relationship between race-specific imprisonment and probation rates in the lower panels of Figure 2. Consider the contrast between Iowa and Massachusetts in the left panel: Iowa has relatively high rates of Black imprisonment and probation, while Massachusetts has low rates of imprisonment and probation. In the right panel, Minnesota reports high probation rates for non-Hispanic Whites but relatively low imprisonment rates, while Oklahoma reports above-average imprisonment rates and below-average probation rates. In addition, states' positioning on the axes of Black imprisonment and probation are not necessarily related to their positioning on the axes of White imprisonment and probation. Some states move very little between the two panels: Oklahoma, for example, has above-average imprisonment rates and below-average probation rates for both populations. Others change quadrants: Iowa shifts from having above-average Black imprisonment and probation rates to being at or below the national average in both White rates.

[FIGURE 3]

When examining (unstandardized) changes over time (Figure 3), variation remains. We also see a clear racial patterning in changing imprisonment rates. Between 2001 and 2017, most states declined in Black imprisonment rates but increased in White imprisonment rates. However, there is no clear relationship between changes in race-specific rates of imprisonment and probation.

Two-Way Fixed Effects Model Results

Main results from fixed-effects models are reported in Figure 4. We display results from Models 2 (top) and 4 (bottom) (the fully-adjusted models for total and race-specific rates, respectively) for only the predictors of interest. Point estimates and standard errors for all predictor and control variables for Models 1-4 appear in Supplementary Information Tables S2-S4.

[FIGURE 4]

Total rates of imprisonment are associated with Black (Figure 4A, Model 2) but not White IMR (Figure 4B, Model 2), such that a one standard-deviation increase in imprisonment rates are associated with 0.65 (*SE: 0.30*) additional infant deaths per 1,000 live births among the Black population only. For context, the Black IMR declined by an average of 0.14 deaths per 1,000 live births per year during the study period, suggesting that a one-standard-deviation increase in imprisonment is comparable to delaying a state's progress in these reductions by more than four years. Total probation and parole rates are not associated with either Black (Figure 4A, Model 2) or White (Figure 4B, Model 2) infant mortality, independent of each other and imprisonment rates. There is no statistically detectable interaction effect between total rates of imprisonment and probation on either Black or White IMRs.

Race-specific rates of imprisonment are not significantly associated with either Black (Figure 4A, Model 4) or White (Figure 4B, Model 4) IMR. Black rates of probation and parole are significantly associated with Black IMR independently of each other and imprisonment rates (Figure 4A, Model 4), such that a one-standard-deviation increase in these predictors results in a decrease in Black IMRs of 0.32 (*SE: 0.12*) and 0.55 (*SE: 0.15*) deaths per 1,000 live births, respectively. Although White rates of imprisonment, probation, and parole are not independently associated with White IMR, the imprisonment x probation term is significant, suggesting that high levels of both rates combine to uniquely and negatively shape White IMR (Figure 4B, Model 4).

Specifically, we find that the impact of probation on White IMR emerges at high rates of imprisonment, such that a one-standard-deviation increase in probation is associated with a larger increase in White IMRs only when imprisonment rates are also high. The estimated coefficient of a one-standard-deviation increase in probation ranges from -.166 (95% CI: -.366 to .034) at the lowest values of imprisonment rates to .458 (95% CI: .149 to .767) at the highest values of imprisonment rates. A visualization of this interaction effect, as well as a histogram displaying the number of cases at each value of imprisonment, can be found in the Supplementary Information.

Discussion and Conclusion

In the US, punishment extends beyond incarceration to community supervision via probation and parole, which together supervise far more people than are held in jail and prison, making it important to understand how all three forms of corrections work in concert to shape population health. Community supervision may mitigate some of the consequences of the carceral state because it is less disruptive to economic and social networks compared to incarceration. At the same time, expansive community supervision efforts may signal heightened control, inducing vigilance and fear that shape the lives of both supervised individuals and their communities. Furthermore, states vary in both *how* they punish (e.g., usage of community supervision versus imprisonment) and *how much* they punish (e.g., levels of community supervision and imprisonment) by race across states and over time (Phelps, 2017; Zimring & Hawkins,

1991), potentially reflecting racialized trajectories of punishment. It is possible that they interact, such that high rates of both indicate particularly expansive surveillance states.

We examine changes in community supervision in US states over a 17-year period and find that increases in Black probation and parole rates, respectively, are associated with decreases in the Black IMR. By contrast, we observe no association between White community supervision and White IMRs. These race-specific results indicate that, for population health, community supervision may serve as a better alternative than incarceration for Black but not White communities. In fact, we find an adverse relation between White probation and White IMRs only when White imprisonment rates also increase. These diverging results between Black and White infants may reflect distinct contextual effects of carceral investment that differentially impact the health of Black and White communities.

The protective effects of community supervision for Black infants, irrespective of imprisonment rates, suggest that community supervision may be less harmful than carceral punishment for Black communities who disproportionately experience the burdens of incarceration. Historically and currently, incarceration in Black communities serves as a form of racialized oppression, leading to disinvestments in education, housing, and health and further concentrating disadvantage and racial segregation in communities (Cadora et al., 2006; Holder et al., 2022). It is possible, then, that reductions in incarceration and increased reliance on probation or parole in Black communities may lessen or even begin to reverse the community burdens of involvement in the criminal legal system in ways that support population health. Our results suggest that this possible “benefit” persists regardless of the level of imprisonment. One speculative explanation is a ceiling effect stemming from persistently high rates of Black imprisonment: Black communities already experience such high levels of imprisonment that any additional (or less) surveillance via imprisonment has little effect on population health, either directly or by mitigating the protective effects of probation (Haskins & Lee, 2016). Alternatively, the null interaction term could reflect the lack of relative variation in Black infant mortality, probation, and incarceration rates jointly, or that almost all states experience a decrease in Black imprisonment making variation on that variable less statistically meaningful.

The deleterious association between probation and White IMRs under conditions of high imprisonment suggests that probation could serve as a net-widener, increasing the proportion of White populations under supervision and compromising community health. The result, if replicated, may underscore how the modern carceral state uses both imprisonment and community supervision to expand its reach and exert control over less commonly targeted communities (e.g., White communities). Indeed, over the past two decades, White supervision rates have increased, while Black supervision rates have declined. Consistent with this pattern of racialized health risk, Patterson (2010) found that the within-prison mortality rate for Black males was considerably lower than the rate for Black males in the general population, while the opposite was true for White males (Patterson, 2010). Researchers suggest that prisons may provide better access to medical care, food, and shelter than the home communities of Black but not White

persons (Massoglia & Pridemore, 2015; Patterson, 2010). Similar patterns emerge in post-prison neighborhood contexts, where Massoglia et al. (2013) found that White but not Black individuals live in significantly more disadvantaged neighborhoods post-prison, noting that, "...because [W]hite offenders generally come from much better neighborhoods, they have much more to lose from a spell of confinement" (p. 142).

Given the relative rarity of imprisonment and social privileges afforded to White (versus Black) communities, contexts characterized by higher levels of both probation and imprisonment may reflect new patterns of disinvestment and disempowerment that partially explain the deleterious effects of justice involvement on infant health for White communities. This lies in contrast to the experiences of Black communities, whom already exist at a relative ceiling of high justice involvement and social harm, such that increasing reliance on forms of supervision that maintain (i.e., probation) or restore (i.e., parole) economic and social ties may appear relatively advantageous.

This paper offers novel insight into the understudied phenomena of community supervision and its potential impact on health disparities in the US. However, some limitations provide avenues for future research. First, while state-level analyses provide valuable perspectives on the impact of the carceral state at the macro-level, most of the daily operations of this system occur at the county-level (Pfaff, 2017). We chose to focus on the state level due to the availability of race-specific estimates of imprisonment, probation, and parole. Future studies, however, could extend parts of our work by using county-level data or lower levels of spatial aggregation. Second, the two-way fixed effect model assumes the absence of time-variant factors that could confound our results. Our models aim to account for potential observable confounders to allow for the most expansive examination of the relationship between supervision and population across both years and states. However, future studies may consider using quasi-experimental methods that may limit the time period or states included in the analysis but account for unobservables. Finally, our findings focus on adult populations under carceral control, which account for the largest portion of the justice-involved population; future research should also consider juvenile populations who constitute a considerable proportion of the carceral population.[3]

Our findings suggest that we can enhance our understanding of the consequences of imprisonment on racial disparities in infant mortality by incorporating both probation and parole into the analytical framework. The results have important theoretical and policy implications for how we should think about probation and parole. Community supervision could be a favorable alternative to imprisonment as a method of correctional control, but it is not a panacea. In disadvantaged communities that have historically experienced high levels of incarceration (e.g., Black communities), community supervision may provide salubrious, benign, or negligible health impacts. In

[3] The population count for individuals under carceral control is even greater when one considers the number of youth who enter the criminal legal system through probation. In 2019, for which complete data is most recently available, 246,000 delinquency cases resulted in youth (juveniles usually aged 10–17) being placed on probation (Hockenberry & Puzzanchera, 2021; McIntyre & Velázquez, 2023).

communities experiencing recent increases in carceral control (e.g., White communities), community supervision may function as a net-widener that may harm the population's health. All of this suggests that community supervision is an intervention that operates within the bounds of the carceral state, and whose consequences require further interrogation.

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Figures and Tables

Table 1 State-level variables, descriptive statistics, and sources

Variable[a]	Mean	SD	Units	Years	Source
Imprisonment Rate	662	246	per 100,000 people	2001-2017	U.S. Bureau of Justice Statistics
Parole Rate	306	260	per 100,000 people	2001-2017	U.S. Bureau of Justice Statistics
Probation Rate	1,829	1,071	per 100,000 people	2001-2017	U.S. Bureau of Justice Statistics
Infant Mortality Rate (NHB)	11.0	2.2	per 1,000 live births	2002-2018	National Vital Statistics System
Infant Mortality Rate (NHW)	5.1	1.1	per 1,000 live births	2002-2018	National Vital Statistics System
High-Risk Birth Index	0.15	0.03	—	2002-2018	National Vital Statistics System
Population Heterogeneity Index	0.19	0.88	—	2002-2018	Various Sources (see Supplementary Information)
Poverty Index	0.20	0.55	—	2002-2018	Various Sources (see Supplementary Information)
Homicide Rate	4.83	2.23	per 100,000 people	2002-2018	Uniform Crime Reports and National Incident-Based Reporting System
Violent Crime Rate	397	149	per 100,000 people	2002-2018	Uniform Crime Reports and National Incident-Based Reporting System
GDP Per Capita	54	10	in 2017 dollars		U.S. Bureau of Economic Analysis
Per Capita Health Expenditure	14,199	3,290	in current dollars		U.S. Center for Medicare and Medicaid Studies

Average SNAP Benefits Per Household	241	45	in current dollars	U.S. Department of Agriculture
Unemployment	5.86	2.02	per 100 people in the labor force	U.S. Bureau of Labor Statistics

[a] We calculate race-specific infant mortality rates based on the maternal state of residence at time of birth and race/ethnicity using the restricted-use birth cohort data. See Supplementary Information Section B.1 and Wildeman (2012) for a detailed description of the high-risk birth, population heterogeneity, and poverty indices. See Supplementary Information Table S1 for more detailed descriptive statistics, including race-specific imprisonment, probation, and parole rates.

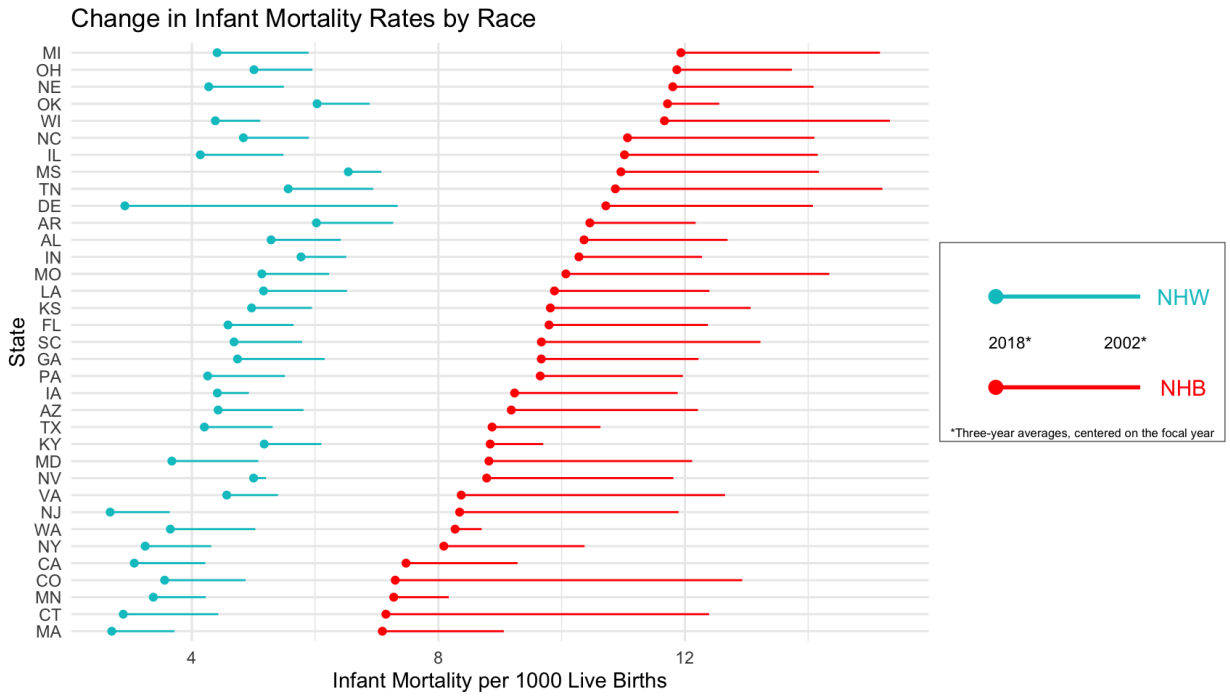


Fig 1 Change in infant mortality rates by race and state, 3-year estimates, 2001-2003 vs. 2017-2019

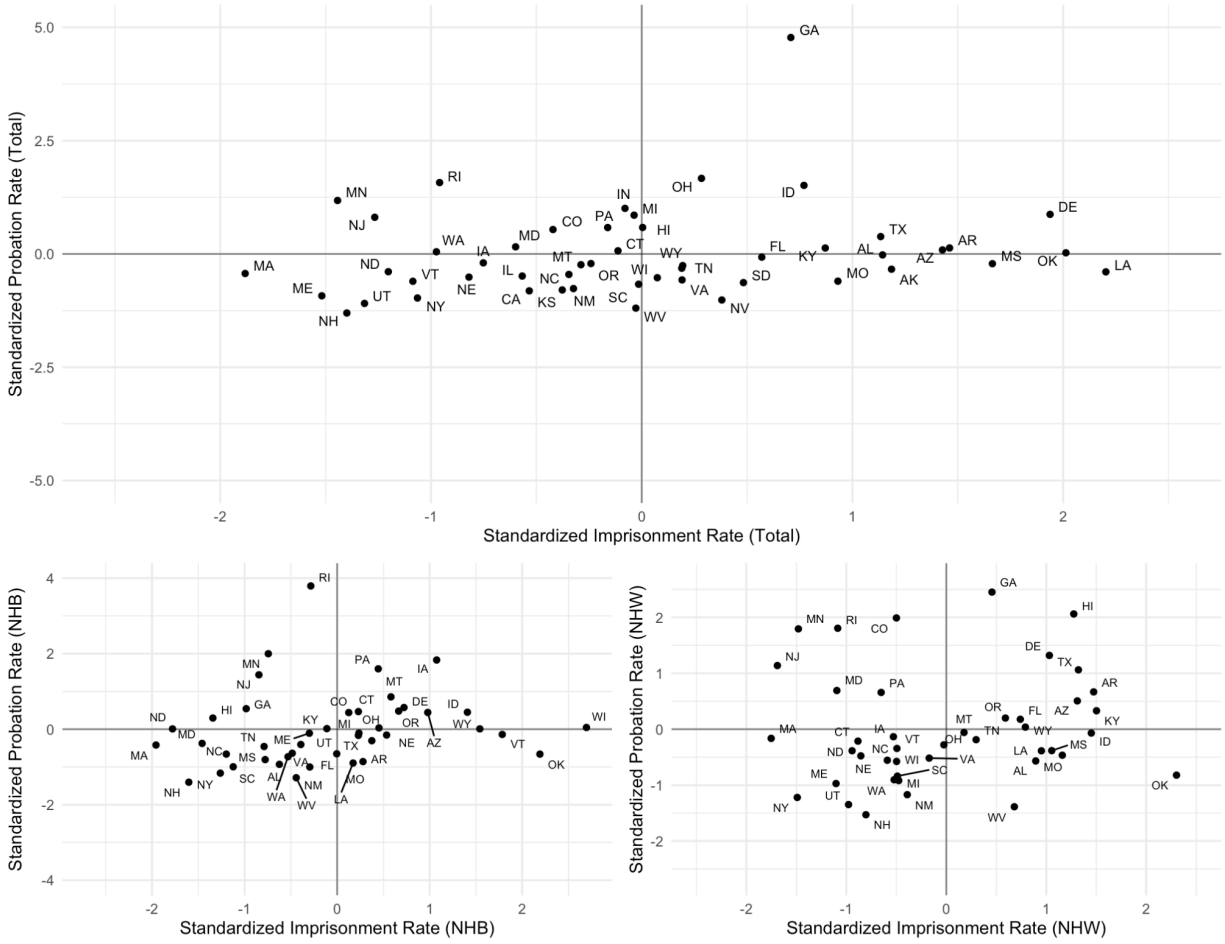


Fig 2 Relationship between imprisonment and probation rates, standardized, total population and non-Hispanic Black and White populations, 2017

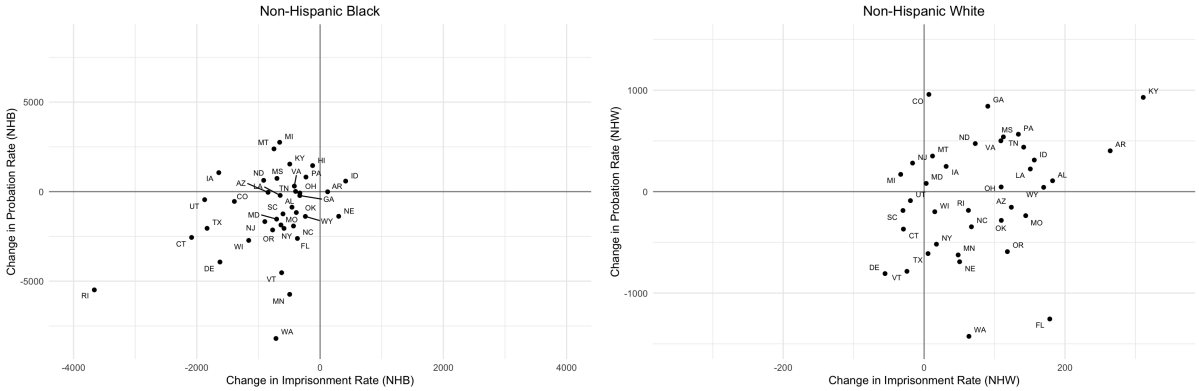


Fig 3 Relationships between change in imprisonment and change in probation rates, by race, between 2001 and 2017

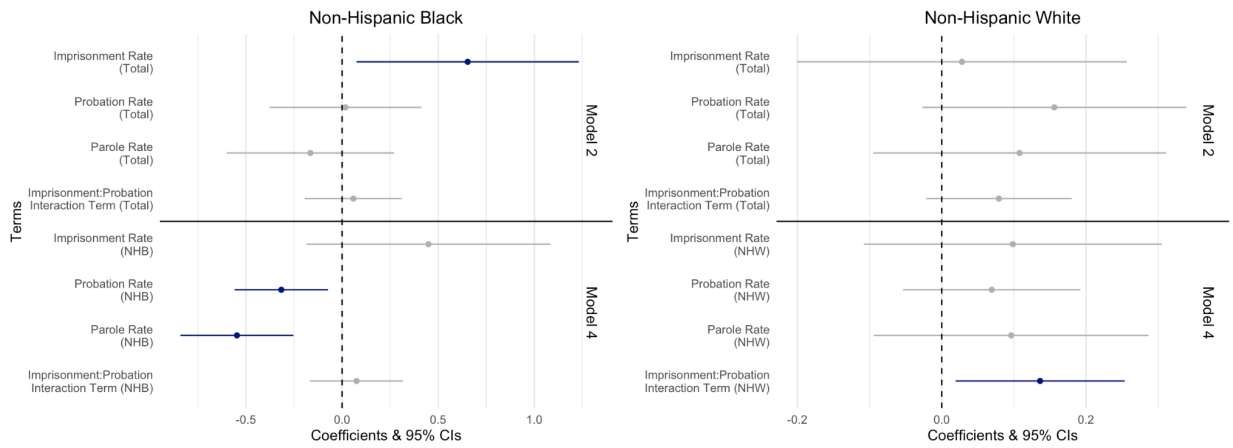


Fig 4 Results from two-way fixed effects models regressing non-Hispanic Black (left) and non-Hispanic White (right) infant mortality rates on total rates of imprisonment, probation, parole (Model 2), and race-specific rates of imprisonment, probation, and parole (Model 4) with contemporaneous controls

SUPPLEMENTARY INFORMATION

State-Level Changes in Community Supervision and Infant Mortality, 2001-2018

Supplementary Information Text

A. Detailed materials.

A.1. Data sources

The Bureau of Justice Statistics provides annual data on state-level probation and parole statistics. The data are collected through the Annual Probation Survey and Annual Parole Survey series, which were downloaded from the Inter-University Consortium for Political and Social Research (ICPSR). Data on individual covariates were collected from various sources that are also publicly available. Specifically, data for the population heterogeneity index was drawn from the Census and American Community Survey (ACS) accessed publicly through the Integrated Public Use Series, USA (IPUMS-USA); from the National Vital Statistics System (NVSS) for the construction of the high-risk birth index; and from the Census and ACS (for the percent of the population that was at least 25 years old with less than a high school education, the percent at or below 138% of the federal poverty threshold, and for the GINI), the Statistical Abstracts of the United States and U.S. Bureau of Labor Statistics (for the percent of the population that was Black), and the U.S. Department of Health and Human Services (for the number of TANF cases per 1,000).

B. Detailed methods.

B.1. Construction of state-level control indexes.

The population heterogeneity index includes: (1) the percent of the population that is foreign born, (2) the percent of the population that is Hispanic, and (3) the percent of the population residing in urban areas ($\alpha = 0.88$). The high-risk birth index includes the per 1,000 live births fetal death rate, and the percent of births that were to unwed mothers, preterm, small for gestational age, and low birthweight ($\alpha = 0.83$). The poverty index includes: (1) the percent of the population that is at least 25 years old with less than a high school education, (2) the percent of the population that is Black, (3) the percent of the population that is at or below 138% of the federal poverty threshold, (4) the GINI, and (5) the number of Temporary Assistance for Needy Families (TANF) cases per 1,000 ($\alpha = 0.73$). All index measures were weighted equally and computed by first standardizing the individual measures, then averaging the z-scores to form the final index.

B.2. Imputation of missing data for index constructions.

In constructing the poverty index, for 2001, there was demographic data on Black resident populations missing from the Statistical Abstracts of the United States, so we substituted data from the Community Population Survey (CPS) accessed publicly through IPUMS CPS. For TANF caseload data, the U.S. Department of Health and Human Services (Office of Family Assistance) only published the total number of TANF recipients, so we took calendar year averages of total recipients and divided them by total resident population numbers from Census estimates to arrive at rates. In constructing the population heterogeneity index, official government data on urban populations are affected by issues of missingness and definitional variation in how the U.S. Census Bureau quantifies urbanicity across census years and 1-year estimates. We attempted to overcome these two problems by implementing a linear spline interpolation of the Decennial Census of Population and Housing (available through the Iowa State University Community Indicators Program: *Urban Percentage of the Population for States*, *Historical*), which recorded urban population percentages from 1900 to 2010. We elected to base our interpolation from the 2000-2010 data points because those years best coincided with our analytic years and shared a stable definition of what the Census Bureau defined as “urban”. We opted for a linear spline interpolation given the steady-state demographic growth observed in urbanicity percentages dating back to the 1900s as opposed to abrupt changes; the large gaps in the missing data; and the broadness of our panel across which we needed to interpolate data.

B.3. Description of all state-level controls

Given the many varied and intersectional pathways that may link imprisonment, probation, and parole to infant mortality at the population level, prior literature has recommended against over-controlling and to limit any controls to a sensible few (Becker et al., 2016; Spector & Brannick, 2011). We therefore only considered plausible measures that were likely associated with imprisonment and community supervision rates, as well as the infant mortality rate. Among relevant controls included were state-level measures of the violent crime and homicide rate, GDP per capita, total public health expenditure per capita, average SNAP benefits per household, and the unemployment rate. Further, since some controls were so highly correlated that including them separately introduced multicollinearity, we followed prior work (Wildeman, 2012) and included a population heterogeneity index, a poverty index, and a high-risk birth index.

Supplementary Information Tables and Figures

Table S1 Descriptive statistics for all variables included in analyses

Variable	Mean (Full Sample)	SD (Full Sample)	Mean (2002)*	SD (2002)*	Mean (2018)*	SD (2018)*
Infant Mortality Rate (NHB)	11.02	2.17	12.54	2.30	9.59	1.60
Infant Mortality Rate (NHW)	5.11	1.10	5.68	0.99	4.51	0.95
Difference in Infant Mortality Rate (NHB-NHW)	5.91	1.82	6.86	2.24	5.08	1.33
Violent Crime Rate	427.07	136.05	478.56	158.56	388.24	114.65
Homicide Rate	5.42	2.09	5.62	2.38	5.44	2.23
State GDP Per Capita	54.29	10.25	49.70	9.31	59.02	11.35
State Public Health Expenditure	14109.33	3164.72	9553.71	1107.91	18486.62	2493.61
Average SNAP Benefit per Family	238.38	38.48	181.36	21.52	244.51	19.29
Unemployment Rate	6.06	2.01	5.55	0.82	3.84	0.57
Population Heterogeneity Index	0.19	0.88	0.01	0.87	0.34	0.90
Poverty Index	0.20	0.55	0.28	0.62	-0.08	0.47
High-Risk Birth Index	0.15	0.03	0.14	0.02	0.15	0.03
Imprisonment Rate (Total)	693.31	243.87	685.57	258.36	659.57	229.99
Imprisonment Rate (NHW)	427.59	178.22	379.00	163.53	448.90	198.27
Imprisonment Rate (NHW)	2451.54	767.52	2752.70	894.43	2094.23	675.63
Probation Rate (Total)	1943.32	1078.27	2007.28	1129.30	1759.96	967.29

Probation Rate (NHW)	1361.28	565.45	1394.16	665.10	1348.79	525.94
Probation Rate (NHB)	3938.34	2014.11	4550.88	2731.68	3376.61	1470.18
Parole Rate (Total)	315.43	265.54	269.62	238.88	341.02	313.53
Parole Rate (NHW)	206.16	194.97	155.18	151.00	244.44	248.45
Parole Rate (NHB)	1063.41	871.31	1046.02	880.27	980.14	914.16

*Imprisonment, probation, and parole rates are lagged by one year, such that the reported values correspond to the years 2001 and 2017.

Table S2 Regression coefficients (standard errors in parentheses) for analysis of Black infant mortality rate, incorporating state and year fixed effects, with and without contemporaneous controls, 2002-2018

	Model 1	Model 2	Model 3	Model 4
Imprisonment Rate (Total)+	0.81 *	0.65 *		
	(0.30)	(0.30)		
Probation Rate (Total)+	-0.10	0.02		
	(0.27)	(0.20)		
Parole Rate (Total)+	-0.26	-0.16		
	(0.18)	(0.22)		
Imprisonment: Probation Interaction Term (Total)+	0.04	0.06		
	(0.13)	(0.13)		
Imprisonment Rate (NHB)+			0.79	0.45
			(0.43)	(0.32)
Probation Rate (NHB)+			-0.42 *	-0.32 *
			(0.19)	(0.12)
Parole Rate (NHB)+			-0.64 ***	-0.55 ***
			(0.16)	(0.15)
Imprisonment: Probation Interaction Term (NHB)+			-0.02	0.08
			(0.13)	(0.12)
Homicide Rate		0.41 *		0.31
		(0.17)		(0.16)
Violent Crime Rate		-0.21		0.07
		(0.19)		(0.22)

GDP Per Capita		1.13 **		1.05 *
		(0.37)		(0.42)
Per Capita Health Expenditure		0.08		-0.20
		(0.47)		(0.42)
Average SNAP Benefit Per Family		-0.89 *		-0.85
		(0.34)		(0.42)
Unemployment Rate		0.24		0.25
		(0.19)		(0.20)
Population Heterogeneity Index		0.84		1.17
		(1.62)		(1.64)
Poverty Index		-0.18		-0.04
		(0.44)		(0.43)
High-Risk Birth Index		0.67		1.74 **
		(0.37)		(0.51)
Num. obs.	576	574	483	481
Num. groups: state	34	34	32	32
Num. groups: year	17	17	17	17
R^2 (full model)	0.68	0.69	0.69	0.71
R^2 (proj. model)	0.02	0.07	0.04	0.10

[+]Predictors are lagged by one year; they range from 2001-2017.
*p<0.05, **p<0.01, ***p<0.001

Table S3 Regression coefficients (standard errors in parentheses) for analysis of White infant mortality rate, incorporating state and year fixed effects, with and without contemporaneous controls, 2002-2018

	Model 1	Model 2	Model 3	Model 4
Imprisonment Rate (Total)[+]	0.12	0.03		
	(0.11)	(0.12)		
Probation Rate (Total)[+]	0.15	0.16		
	(0.08)	(0.09)		
Parole Rate (Total)[+]	0.09	0.11		
	(0.08)	(0.10)		
Imprisonment: Probation Interaction Term (Total)[+]	0.09	0.08		
	(0.05)	(0.05)		
Imprisonment Rate (NHW)[+]			0.08	0.10
			(0.12)	(0.11)

Probation Rate (NHW)[+]			0.04	0.07
			(0.05)	(0.06)
Parole Rate (NHW)[+]			0.11	0.10
			(0.09)	(0.10)
Imprisonment: Probation Interaction Term (NHW)			0.14 *	0.14 *
			(0.06)	(0.06)
Homicide Rate		0.13		0.13
		(0.08)		(0.09)
Violent Crime Rate		0.08		0.13
		(0.07)		(0.09)
GDP Per Capita		0.28		0.33
		(0.20)		(0.26)
Per Capita Health Expenditure		-0.06		-0.18
		(0.14)		(0.15)
Average SNAP Benefit Per Family		0.06		0.12
		(0.10)		(0.11)
Unemployment Rate		0.03		-0.02
		(0.07)		(0.07)
Population Heterogeneity Index		0.31		0.39
		(0.39)		(0.51)
Poverty Index		-0.05		-0.07
		(0.11)		(0.14)
High-Risk Birth Index		0.16		0.06
		(0.11)		(0.24)
Num. obs.	576	574	483	481
Num. groups: state	34	34	32	32
Num. groups: year	17	17	17	17
R^2 (full model)	0.87	0.87	0.86	0.87
R^2 (proj. model)	0.03	0.06	0.03	0.09

[+]Predictors are lagged by one year; they range from 2001-2017.
*p<0.05, **p<0.01, ***p<0.001

Table S4 Regression coefficients (standard errors in parentheses) for analysis of the difference in Black and White infant mortality rates, incorporating state and year fixed effects, with and without contemporaneous controls, 2002-2018

	Model 1	Model 2	Model 3	Model 4
Imprisonment Rate (Total)[+]	0.69 *	0.63 *		
	(0.29)	(0.26)		
Probation Rate (Total)[+]	-0.25	-0.14		
	(0.27)	(0.20)		
Parole Rate (Total)[+]	-0.35	-0.27		
	(0.19)	(0.21)		
Imprisonment: Probation Interaction Term (Total)[+]	-0.05	-0.02		
	(0.10)	(0.10)		
Imprisonment Rate (NHB-NHW)[+]			0.56	0.36
			(0.43)	(0.37)
Probation Rate (NHB-NHW)[+]			-0.39 *	-0.31 *
			(0.17)	(0.12)
Parole Rate (NHB-NHW)[+]			-0.74 ***	-0.61 **
			(0.19)	(0.17)
Imprisonment: Probation Interaction Term (NHB-NHW)			0.02	0.06
			(0.11)	(0.11)
Homicide Rate		0.28		0.21
		(0.21)		(0.22)
Violent Crime Rate		-0.29		-0.14
		(0.22)		(0.28)
GDP Per Capita		0.85		0.82
		(0.46)		(0.63)
Per Capita Health Expenditure		0.14		-0.06
		(0.46)		(0.45)
Average SNAP Benefit Per Family		-0.94 **		-0.94 *
		(0.33)		(0.43)
Unemployment Rate		0.21		0.26
		(0.22)		(0.24)
Population Heterogeneity Index		0.53		0.94
		(1.69)		(1.79)
Poverty Index		-0.13		0.09

		(0.44)		(0.43)
High-Risk Birth Index		0.51		1.68 *
		(0.36)		(0.66)
Num. obs.	576	574	483	481
Num. groups: state	34	34	32	32
Num. groups: year	17	17	17	17
R^2 (full model)	0.48	0.50	0.52	0.54
R^2 (proj. model)	0.01	0.05	0.03	0.08

[+]Predictors are lagged by one year; they range from 2001-2017.
*p<0.05, **p<0.01, ***p<0.001

Table S5 Single-predictor models predicting Black infant mortality, with fixed effects for state and year and contemporaneous controls (standard errors in parentheses), 2002-2018

	Model 1	Model 2	Model 3
Imprisonment Rate (NHB)[+]	0.30		
	(0.32)		
Probation Rate (NHB)[+]		-0.17	
		(0.12)	
Parole Rate (NHB)[+]			-0.47 **
			(0.14)
Homicide Rate	0.40 *	0.31	0.28
	(0.17)	(0.17)	(0.16)
Violent Crime Rate	-0.12	-0.05	0.07
	(0.19)	(0.20)	(0.19)
GDP Per Capita	1.21 **	1.42 **	1.08 **
	(0.40)	(0.40)	(0.37)
Per Capita Health Expenditure	0.01	-0.42	-0.08
	(0.48)	(0.36)	(0.42)
Average SNAP Benefit Per Family	-0.88 *	-0.90 *	-0.89 *
	(0.34)	(0.42)	(0.37)
Unemployment Rate	0.27	0.31	0.39
	(0.19)	(0.20)	(0.19)
Population Heterogeneity Index	0.57	1.10	1.86
	(1.59)	(1.55)	(1.58)
Poverty Index	-0.19	-0.08	-0.24
	(0.43)	(0.46)	(0.45)

High-Risk Birth Index	0.73	1.97 **	0.93 *
	(0.37)	(0.56)	(0.42)
Num. obs.	576	484	549
Num. groups: state	34	32	34
Num. groups: year	17	17	17
R^2 (full model)	0.69	0.70	0.70
R^2 (proj. model)	0.06	0.08	0.06

[+]Predictors are lagged by one year; they range from 2001-2017.
*p<0.05, **p<0.01, ***p<0.001

Table S6 Single-predictor models predicting White infant mortality, with fixed effects for state and year and contemporaneous controls (standard errors in parentheses), 2002-2018

	Model 1	Model 2	Model 3
Imprisonment Rate (NHW)[+]	0.02		
	(0.11)		
Probation Rate (NHW)[+]		0.04	
		(0.06)	
Parole Rate (NHW)[+]			0.14
			(0.10)
Homicide Rate	0.11	0.12	0.12
	(0.07)	(0.09)	(0.08)
Violent Crime Rate	0.13 *	0.17	0.10
	(0.06)	(0.10)	(0.09)
GDP Per Capita	0.28	0.29	0.36
	(0.22)	(0.29)	(0.24)
Per Capita Health Expenditure	-0.11	-0.16	-0.06
	(0.18)	(0.16)	(0.15)
Average SNAP Benefit Per Family	0.05	0.12	0.12
	(0.11)	(0.12)	(0.11)
Unemployment Rate	0.04	0.00	0.02
	(0.06)	(0.07)	(0.07)
Population Heterogeneity Index	0.02	0.31	-0.13
	(0.42)	(0.53)	(0.50)
Poverty Index	-0.09	-0.12	-0.10
	(0.11)	(0.15)	(0.11)
High-Risk Birth Index	0.19	0.08	0.12

	(0.12)	(0.26)	(0.12)
Num. obs.	576	484	549
Num. groups: state	34	32	34
Num. groups: year	17	17	17
R^2 (full model)	0.87	0.86	0.87
R^2 (proj. model)	0.05	0.06	0.07

[+]Predictors are lagged by one year; they range from 2001-2017.
*p<0.05, **p<0.01, ***p<0.001

Table S7 Single-predictor models predicting the difference between Black and White infant mortality, with fixed effects for state and year and contemporaneous controls (standard errors in parentheses), 2002-2018

	Model 1	Model 2	Model 3
Imprisonment Rate (NHB-NHW)[+]	0.17		
	(0.32)		
Probation Rate (NHB-NHW)[+]		-0.13	
		(0.12)	
Parole Rate (NHB-NHW)[+]			-0.54 **
			(0.16)
Homicide Rate	0.29	0.21	0.18
	(0.21)	(0.23)	(0.22)
Violent Crime Rate	-0.26	-0.23	-0.08
	(0.21)	(0.27)	(0.22)
GDP Per Capita	0.96	1.16	0.76
	(0.50)	(0.59)	(0.49)
Per Capita Health Expenditure	0.09	-0.25	-0.03
	(0.49)	(0.38)	(0.40)
Average SNAP Benefit Per Family	-0.94 **	-1.03 *	-0.97 *
	(0.34)	(0.43)	(0.37)
Unemployment Rate	0.25	0.30	0.35
	(0.22)	(0.24)	(0.23)
Population Heterogeneity Index	0.54	0.99	2.01
	(1.70)	(1.83)	(1.70)
Poverty Index	-0.13	0.07	-0.08
	(0.44)	(0.44)	(0.44)
High-Risk Birth Index	0.58	1.90 *	0.76
	(0.37)	(0.70)	(0.46)

Num. obs.	576	484	549
Num. groups: state	34	32	34
Num. groups: year	17	17	17
R^2 (full model)	0.50	0.53	0.51
R^2 (proj. model)	0.04	0.06	0.05

[+]Predictors are lagged by one year; they range from 2001-2017.
*p<0.05, **p<0.01, ***p<0.001

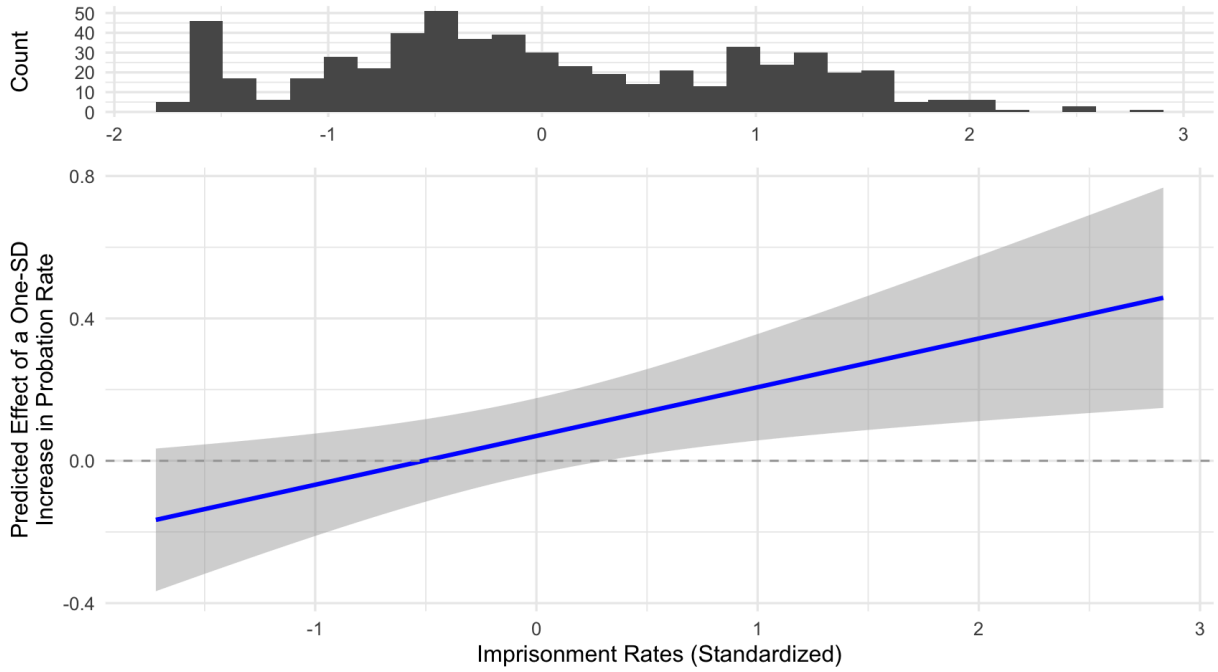


Fig S1 Number of cases and predicted effect of a one-standard-deviation increase in probation rate on White infant mortality rates at various imprisonment rates

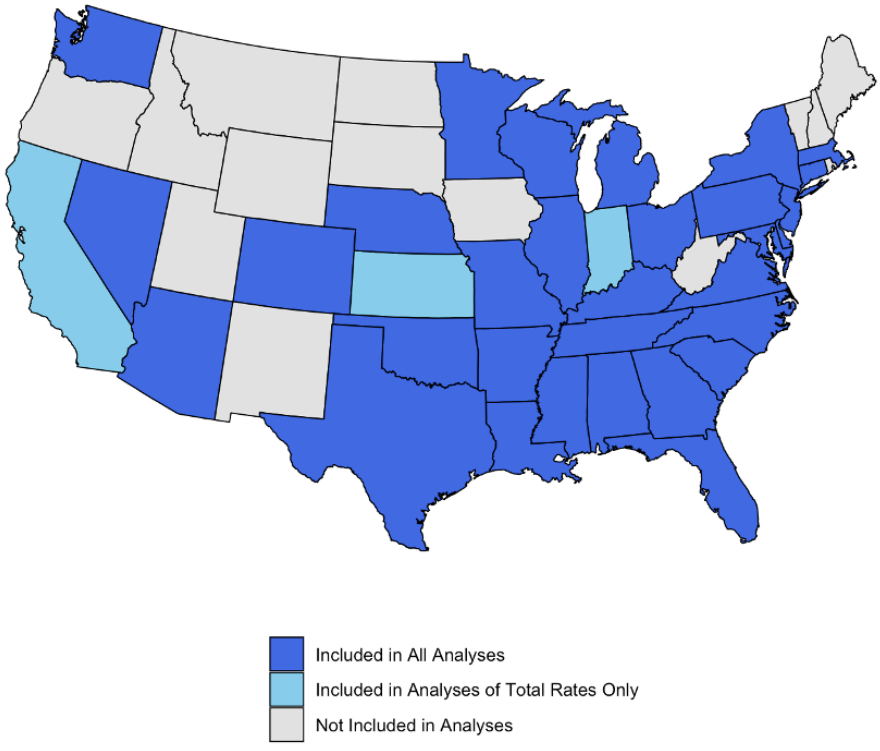


Fig S2 Map of included states in analysis, by analysis type

Supplementary Information References

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